Heart Disease: A Comprehensive Reference

An educational reference covering coronary artery disease, heart failure, arrhythmias, valvular disease, diagnosis, treatment, and prevention. Compiled for educational and informational use only. This document is not medical advice.

1. Overview of Cardiovascular Disease

Cardiovascular disease (CVD) is an umbrella term for disorders of the heart and blood vessels. It is the leading cause of death worldwide, responsible for an estimated 17.9 million deaths each year according to the World Health Organization. The term encompasses coronary artery disease (CAD), heart failure (HF), arrhythmias, valvular heart disease, congenital heart disease, peripheral artery disease, and cerebrovascular disease.

The heart is a four-chambered muscular pump consisting of two atria and two ventricles. The right side of the heart receives deoxygenated blood from the body and pumps it to the lungs through the pulmonary artery. The left side receives oxygenated blood from the lungs via the pulmonary veins and pumps it to the rest of the body through the aorta. Coordinated contraction is driven by the cardiac conduction system: the sinoatrial (SA) node, atrioventricular (AV) node, bundle of His, and Purkinje fibers.

1.1 Major Categories

Coronary artery disease (CAD) involves the narrowing or blockage of coronary arteries, typically by atherosclerotic plaque. Heart failure (HF) is the inability of the heart to pump sufficient blood to meet the body's metabolic demands. Arrhythmias are disturbances in the rate or rhythm of the heartbeat, including atrial fibrillation, ventricular tachycardia, and bradyarrhythmias. Valvular heart disease includes mitral regurgitation, aortic stenosis, and other disorders of the four cardiac valves. Congenital heart disease comprises structural defects present from birth, such as atrial septal defect (ASD) and tetralogy of Fallot.

2. Coronary Artery Disease (CAD)

Coronary artery disease results from atherosclerosis, a chronic inflammatory process in which lipid-laden plaques accumulate in the intima of the coronary arteries. Over years to decades, these plaques narrow the arterial lumen and reduce myocardial blood flow. Plaque rupture or erosion can trigger thrombus formation, leading to acute coronary syndromes (ACS) including unstable angina, non-ST-elevation myocardial infarction (NSTEMI), and ST-elevation myocardial infarction (STEMI).

2.1 Symptoms

Stable angina presents as substernal chest pressure or tightness, typically provoked by exertion or emotional stress and relieved by rest or sublingual nitroglycerin within 5 minutes. Symptoms may radiate to the left arm, jaw, neck, or back. Women, elderly patients, and those with diabetes more often experience atypical presentations such as fatigue, dyspnea, nausea, or epigastric discomfort. Acute myocardial infarction is suggested by chest pain lasting more than 20 minutes, unrelieved by nitroglycerin, often accompanied by diaphoresis, nausea, and a sense of impending doom.

2.2 Diagnosis

Initial workup includes a 12-lead electrocardiogram (ECG), serum cardiac troponin (preferably high-sensitivity troponin T or I), and a focused history and physical examination. ST-segment elevation on ECG in two contiguous leads with elevated troponin defines STEMI. Stable patients may undergo stress testing, coronary CT angiography (CCTA), or invasive coronary angiography. The Diamond-Forrester score and pretest probability assessment guide test selection.

2.3 Treatment

STEMI requires emergent reperfusion. Primary percutaneous coronary intervention (PCI) is preferred when available within 90 minutes of first medical contact (door-to-balloon time). Fibrinolytic therapy with agents such as tenecteplase or alteplase is an alternative when timely PCI is not feasible. NSTEMI and unstable angina are managed with dual antiplatelet therapy (aspirin plus a P2Y12 inhibitor such as clopidogrel, ticagrelor, or prasugrel), anticoagulation with heparin, and risk-stratified invasive strategy.

Long-term secondary prevention includes high-intensity statin therapy (atorvastatin 40 to 80 mg or rosuvastatin 20 to 40 mg), aspirin 81 mg daily, beta-blockers, and ACE inhibitors or ARBs in patients with reduced ejection fraction or hypertension. Lifestyle modification, including smoking cessation, structured exercise, and a Mediterranean-style diet, is essential.

3. Heart Failure

Heart failure (HF) is a clinical syndrome resulting from any structural or functional impairment of ventricular filling or ejection of blood. It affects approximately 6.2 million adults in the United States and is a leading cause of hospitalization in patients over age 65. Heart failure is classified by left ventricular ejection fraction (LVEF) into three categories.

3.1 Classification by Ejection Fraction

Category	LVEF Range	Common Causes
HFrEF (reduced)	≤ 40%	Ischemic cardiomyopathy, dilated cardiomyopathy, prior MI
HFmrEF (mildly reduced)	41 to 49%	Transitional state, partial recovery
HFpEF (preserved)	≥ 50%	Hypertension, diabetes, obesity, aging, infiltrative disease

3.2 NYHA Functional Classification

The New York Heart Association (NYHA) classification grades symptom severity. Class I patients have no limitation of physical activity. Class II have slight limitation; ordinary activity causes symptoms. Class III have marked limitation; less than ordinary activity causes symptoms. Class IV patients have symptoms at rest and are unable to carry out any physical activity without discomfort.

3.3 Symptoms and Signs

Common symptoms include dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea (PND), fatigue, and peripheral edema. Physical examination findings may include elevated jugular venous pressure (JVP), an S3 gallop, bibasilar pulmonary crackles, hepatomegaly, and pitting edema of the lower extremities. The presence of a displaced apical impulse may indicate cardiomegaly.

3.4 Diagnosis

Initial evaluation includes B-type natriuretic peptide (BNP) or N-terminal pro-BNP (NT-proBNP), 12-lead ECG, chest X-ray, and transthoracic echocardiography (TTE). BNP greater than 100 pg/mL or NT-proBNP greater than 300 pg/mL supports the diagnosis in the acute setting. Echocardiography quantifies LVEF, chamber dimensions, valve function, and diastolic parameters. Cardiac MRI may be used when echocardiographic windows are inadequate or to characterize myocardial tissue.

3.5 Guideline-Directed Medical Therapy (GDMT) for HFrEF

Modern HFrEF therapy is built on four pillars, often called the 'four pillars of HF therapy': (1) Angiotensin receptor-neprilysin inhibitor (ARNI) such as sacubitril/valsartan, or alternatively an ACE inhibitor or ARB; (2) Evidence-based beta-blocker (carvedilol, metoprolol succinate, or bisoprolol); (3) Mineralocorticoid receptor antagonist (spironolactone or eplerenone); and (4) SGLT2 inhibitor (dapagliflozin or empagliflozin). All four pillars should be initiated and uptitrated as tolerated, regardless of diabetes status.

Adjunctive therapies include loop diuretics (furosemide, torsemide, bumetanide) for volume overload, ivabradine for selected patients with heart rate above 70 bpm in sinus rhythm, hydralazine plus isosorbide dinitrate (particularly in self-identified Black patients), and device therapies including implantable cardioverter defibrillators (ICDs) and cardiac resynchronization therapy (CRT) when indicated.

4. Arrhythmias

Arrhythmias are abnormalities in the rate, rhythm, or conduction of the heartbeat. They range from benign premature atrial contractions to life-threatening ventricular fibrillation. Mechanisms include reentry, abnormal automaticity, and triggered activity.

4.1 Atrial Fibrillation

Atrial fibrillation (AF) is the most common sustained arrhythmia, characterized by irregularly irregular atrial activation and absence of distinct P waves on ECG. Prevalence increases with age, affecting roughly 9% of individuals over 65. Management follows three pillars: rate control (beta-blockers, non-dihydropyridine calcium channel blockers, or digoxin), rhythm control (antiarrhythmic drugs such as amiodarone, flecainide, or catheter ablation), and stroke prevention with anticoagulation. Anticoagulation decisions are guided by the CHA2DS2-VASc score; a score of 2 or more in men or 3 or more in women generally warrants oral anticoagulation with a direct oral anticoagulant (DOAC) such as apixaban, rivaroxaban, edoxaban, or dabigatran, or with warfarin (target INR 2.0 to 3.0).

4.2 Ventricular Arrhythmias

Ventricular tachycardia (VT) is defined as three or more consecutive ventricular beats at a rate above 100 bpm. Sustained VT lasting more than 30 seconds or causing hemodynamic instability is a medical emergency. Ventricular fibrillation (VF) is characterized by chaotic electrical activity and absence of effective cardiac output, requiring immediate defibrillation. Patients with prior cardiac arrest, sustained VT, or LVEF $\leq$ 35% on optimal medical therapy may be candidates for an implantable cardioverter defibrillator (ICD) for primary or secondary prevention of sudden cardiac death.

4.3 Bradyarrhythmias

Bradyarrhythmias include sinus node dysfunction (sick sinus syndrome) and atrioventricular (AV) blocks. First-degree AV block features a PR interval greater than 200 ms. Second-degree AV block is subdivided into Mobitz Type I (Wenckebach), with progressive PR prolongation until a beat is dropped, and Mobitz Type II, with intermittent non-conducted P waves and a fixed PR interval. Third-degree (complete) AV block represents complete dissociation between atrial and ventricular activity. Symptomatic bradyarrhythmias generally require permanent pacemaker implantation.

5. Valvular Heart Disease

Valvular heart disease results from stenosis (narrowing) or regurgitation (leaking) of the four cardiac valves: mitral, aortic, tricuspid, and pulmonic. Severity is graded by echocardiography using parameters such as transvalvular pressure gradients, valve area, and regurgitant volume.

5.1 Aortic Stenosis

Aortic stenosis is most commonly caused by age-related calcific degeneration of a tricuspid aortic valve or by a congenital bicuspid aortic valve. Severe AS is defined by an aortic valve area less than 1.0 cm^2, peak transvalvular velocity greater than 4.0 m/s, or mean gradient above 40 mmHg. Classic symptoms are angina, syncope, and dyspnea (the 'SAD triad'); onset of symptoms portends a poor prognosis without intervention. Treatment options include surgical aortic valve replacement (SAVR) and transcatheter aortic valve replacement (TAVR), with TAVR increasingly used across all surgical risk strata.

5.2 Mitral Regurgitation

Mitral regurgitation (MR) may be primary (degenerative, due to leaflet pathology such as mitral valve prolapse) or secondary (functional, due to left ventricular dilatation or papillary muscle dysfunction). Severe MR is characterized by a regurgitant volume of 60 mL or more per beat, regurgitant fraction above 50%, and effective regurgitant orifice area (EROA) of 0.40 cm^2 or greater. Surgical repair is preferred over replacement when feasible. Transcatheter edge-to-edge repair (TEER) with the MitraClip device is an option for selected patients with prohibitive surgical risk.

6. Risk Factors and Prevention

Cardiovascular risk factors are categorized as non-modifiable and modifiable. Non-modifiable factors include age (men ≥ 45, women ≥ 55), male sex, family history of premature CAD (first-degree male relative under 55 or female relative under 65), and ethnicity. Modifiable factors include hypertension, dyslipidemia, diabetes mellitus, tobacco use, obesity, physical inactivity, and unhealthy diet.

6.1 Blood Pressure Targets

Category	Systolic (mmHg)	Diastolic (mmHg)
Normal	< 120	< 80
Elevated	120 to 129	< 80
Stage 1 Hypertension	130 to 139	80 to 89
Stage 2 Hypertension	≥ 140	≥ 90
Hypertensive Crisis	> 180	> 120

Per 2017 ACC/AHA guidelines, hypertension is diagnosed at 130/80 mmHg or higher. The general target for adults with hypertension and elevated cardiovascular risk is below 130/80 mmHg. First-line agents include thiazide diuretics, ACE inhibitors, ARBs, and dihydropyridine calcium channel blockers.

6.2 Lipid Targets

LDL cholesterol is the primary target of lipid-lowering therapy. For patients with established atherosclerotic cardiovascular disease (ASCVD), the goal is LDL-C less than 70 mg/dL, and below 55 mg/dL in very high-risk patients per ESC guidelines. High-intensity statins reduce LDL-C by 50% or more. PCSK9 inhibitors (evolocumab, alirocumab) and ezetimibe are added when statin therapy is insufficient. Inclisiran, an siRNA targeting PCSK9, is administered subcutaneously every 6 months.

6.3 Diabetes and Cardiovascular Risk

Diabetes mellitus is a CAD risk equivalent. HbA1c targets are individualized; a goal of less than 7% is typical for most adults. SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) and GLP-1 receptor agonists (semaglutide, liraglutide, dulaglutide) reduce major adverse cardiovascular events independent of glucose lowering and are recommended in patients with type 2 diabetes and established or high risk of ASCVD.

6.4 Lifestyle Recommendations

Lifestyle modification is foundational. Recommendations include at least 150 minutes per week of moderate-intensity aerobic activity or 75 minutes of vigorous activity, plus muscle-strengthening on 2 or more days per week. A Mediterranean or DASH dietary pattern is preferred, emphasizing vegetables, fruits, whole grains, legumes, nuts, olive oil, and fish, while limiting red and processed meats, refined grains, and added sugars. Sodium intake should be limited to less than 2,300 mg per day, ideally 1,500 mg per day. Tobacco cessation is the single most impactful intervention; smoking cessation reduces CAD mortality by approximately 36%.

7. Diagnostic Tests

7.1 Electrocardiography (ECG)

The 12-lead ECG records cardiac electrical activity from 12 vectors. Normal sinus rhythm is characterized by a rate between 60 and 100 bpm with a P wave preceding each QRS complex. The PR interval normally ranges from 120 to 200 ms; the QRS duration is less than 120 ms; the corrected QT interval (QTc) is below 460 ms in women and below 450 ms in men. ST-segment elevation, T-wave inversion, and pathological Q waves suggest myocardial ischemia or infarction.

7.2 Echocardiography

Transthoracic echocardiography (TTE) is the workhorse of cardiac imaging. It provides assessment of chamber size, wall thickness, systolic and diastolic function, and valvular structure. Transesophageal echocardiography (TEE) offers superior visualization of the left atrial appendage, prosthetic valves, and aortic dissection. Stress echocardiography (exercise or pharmacologic with dobutamine) detects inducible wall motion abnormalities indicative of ischemia.

7.3 Cardiac Biomarkers

High-sensitivity cardiac troponin (hs-cTnT or hs-cTnI) is the preferred biomarker for myocardial injury. Levels above the 99th percentile of the upper reference limit, with a rising or falling pattern, support the diagnosis of acute myocardial infarction. BNP and NT-proBNP are markers of myocardial wall stress and are useful in diagnosing and prognosticating heart failure. CK-MB is now rarely used as a stand-alone marker due to lower specificity.

7.4 Advanced Imaging

Coronary CT angiography (CCTA) provides non-invasive visualization of the coronary arteries and is recommended for symptomatic patients at low to intermediate pretest probability of CAD. Invasive coronary angiography remains the gold standard and enables ad hoc PCI. Cardiac MRI characterizes myocardial tissue (e.g., late gadolinium enhancement for fibrosis), assesses ventricular function with high accuracy, and is useful in myocarditis, infiltrative cardiomyopathies, and congenital heart disease.

8. Special Topics

8.1 Cardiogenic Shock

Cardiogenic shock is a state of end-organ hypoperfusion due to primary cardiac dysfunction, with mortality rates historically near 50%. It is defined by systolic blood pressure below 90 mmHg for more than 30 minutes (or need for vasopressors), evidence of pulmonary congestion, and signs of impaired organ perfusion. Management includes early revascularization for ischemic etiology, inotropic support (dobutamine, milrinone), vasopressors (norepinephrine), and mechanical circulatory support such as intra-aortic balloon pump (IABP), Impella, veno-arterial extracorporeal membrane oxygenation (VA-ECMO), or durable left ventricular assist device (LVAD).

8.2 Pericardial Disease

Acute pericarditis presents with pleuritic chest pain that improves leaning forward, a friction rub on auscultation, diffuse ST elevation and PR depression on ECG, and possible pericardial effusion. Treatment is with NSAIDs and colchicine for 3 months. Cardiac tamponade is a clinical diagnosis featuring Beck's triad: hypotension, muffled heart sounds, and elevated JVP, plus pulsus paradoxus. Emergent pericardiocentesis is lifesaving.

8.3 Cardiomyopathies

Hypertrophic cardiomyopathy (HCM) is a genetic disorder of the sarcomere causing left ventricular hypertrophy in the absence of loading conditions. Prevalence is approximately 1 in 500. Symptoms include dyspnea, chest pain, syncope, and palpitations; HCM is a leading cause of sudden cardiac death in young athletes. Dilated cardiomyopathy involves chamber dilation with systolic dysfunction; etiologies include ischemic disease, alcohol, peripartum, viral myocarditis, and genetic mutations. Restrictive cardiomyopathy features impaired ventricular filling with preserved or near-normal systolic function and may result from amyloidosis, sarcoidosis, or hemochromatosis.

8.4 Sex Differences and Pregnancy

Women with acute coronary syndrome more often present with atypical symptoms such as fatigue, dyspnea, and back or jaw pain. Spontaneous coronary artery dissection (SCAD) is a non-atherosclerotic cause of MI, disproportionately affecting younger women. Pregnancy is associated with increased plasma volume and cardiac output, which can unmask underlying cardiomyopathy. Peripartum cardiomyopathy is defined by HF developing in the last month of pregnancy or within 5 months postpartum without other identifiable cause.

8.5 When to Seek Emergency Care

Patients should call emergency services immediately for chest pain or pressure lasting more than a few minutes, particularly when associated with shortness of breath, sweating, nausea, lightheadedness, or pain radiating to the arm, jaw, or back. Other emergencies include sudden severe shortness of breath, syncope, palpitations with chest pain, and signs of stroke (sudden weakness, facial droop, slurred speech).